Thus, although there is wide agreement that environment-induced decoherence is a crucial part of the structure spanning the quantum-to-classical divide, and while many are hopeful that these considerations will one day coalesce into a complete and cogent connection between the two, far from everyone is convinced that the bridge has yet been fully built.

Quantum Mechanics and the Arrow of Time

So where do we stand on the measurement problem, and what does it mean for the arrow of time? Broadly speaking, there are two classes of proposals for linking common experience with quantum reality. In the first class (for example, wavefunction as knowledge; Many Worlds; decoherence), Schrödinger's equation is the be-all and end-all of the story; the proposals simply provide different ways of interpreting what the equation means for physical reality. In the second class (for example, Bohm; Ghirardi-Rimini-Weber), Schrödinger's equation must be supplemented with other equations (in Bohm's case, an equation that shows how a wavefunction pushes a particle around) or it must be modified (in the Ghirardi-Rimini-Weber case, to incorporate a new, explicit collapse mechanism). A key question for determining the impact on time's arrow is whether these proposals introduce a fundamental asymmetry between one direction in time and the other. Remember, Schrödinger's equation, just like those of Newton, Maxwell, and Einstein, treats forward and backward in time on a completely equal footing. It provides no arrow to temporal evolution. Do any of the proposals change this?

In the first class of proposals, the Schrödinger framework is not at all modified, so temporal symmetry is maintained. In the second class, temporal symmetry may or may not survive, depending on the details. For example, in Bohm's approach, the new equation proposed does treat time future and time past on an equal footing and so no asymmetry is introduced. However, the proposal of Ghirardi, Rimini, and Weber introduces a collapse mechanism that *does* have a temporal arrow—an "uncollapsing" wavefunction, one that goes from a spiked to a spread-out shape, would not conform to the modified equations. Thus, depending on the proposal, quantum mechanics, together with a